
Data Card

Optimizing RAG Document Retrieval for Agronomic Advice

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Input vs. Output

Input: A farmer’s natural-language question (e.g., “Why are my maize leaves turning yellow?”).

Output: A ranked list of the top-5 most relevant document IDs from a knowledge base of 695 agricultural extension factsheets, ordered from most to least relevant.

Classification Problem

Not a classification task in the traditional sense. This is a **ranking & retrieval task**. Rather than predicting discrete labels, the model scores and ranks candidate documents by relevance to a query. Relevance itself is graded on a 0–3 scale:

- **0** – Not relevant
- **1** – Marginal / different crop
- **2** – Related aspect
- **3** – Perfect answer

This functions similarly to an ordinal classification of relevance per document.

Labelling

Relevance labels (qrels) were provided by the competition organisers, derived from expert judgement on which documents genuinely answer each training query. We did not create these labels ourselves; we rely on the pre-supplied `qrels_train.csv` for local evaluation and trust the organiser’s hidden expert judgements for the true evaluation set.

We check consistency by using the provided `evaluate_ndcg_at_5` function to validate our own retrieval outputs against the known training labels before submitting.

Data Creation

The document corpus (695 factsheets) and query sets were provided directly by the competition organisers—we are not collecting or scraping this data ourselves. Per the brief, the corpus mixes original factsheets with rewrites of real CC-BY-licensed agricultural research, meaning it



is derived from legitimately licensed real-world agricultural knowledge rather than synthetic or scraped-without-permission content.

Web Collection Ethics

We are not independently sourcing data from the web for this project. The underlying corpus was already curated by the organisers from CC-BY-licensed agricultural research, which permits reuse with attribution. We are not introducing new web-scraped data, so no additional consent, copyright, or scraping-ethics concerns arise from our side of the work.

Bias and Representation

The dataset is described as covering crop diseases, pests, nutrient deficiencies, soil management, climate adaptation, and fertiliser advice for Sub-Saharan African agriculture—but we do not know the exact crop/regional distribution across the 695 factsheets.

Crops or regions with less documented research (e.g., minor/local crop varieties, or regions with fewer formal extension resources) may be under-represented, meaning our retrieval system could perform less reliably for farmers growing those crops.

We plan to check this by reviewing which crops and topics appear in our lowest-nDCG@5 training queries, to see if certain crops are systematically harder to retrieve for.

Values and Community Alignment

Our dataset design supports the values identified in Challenge 1—accuracy, accessibility, and trustworthiness—because working with a real, expert-labelled dataset, rather than one we scraped or synthesised ourselves, ensures our evaluation reflects genuine expert judgement about what farmers actually need.

We remain accountable to the intended community by being transparent about where the dataset’s coverage may be uneven, and by prioritising retrieval quality on hard negatives (documents that look similar but are wrong) rather than optimising only for leaderboard performance.